

Dendogramas

Equipo 9

2025-09-07

###WDBCclustering

Conjunto de datos: `loan_approved_dataset`. Es una base de datos de clientes de instituciones financieras.

Dimensión de la base de datos

```
df %>% dim()
```

```
## [1] 4269 12
```

La base contiene 12 variables, incluida una variable respuesta. Son 4269 renglones.

Hacemos un pequeño descriptivo, ya se hizo en el proyecto anterior.

```
df %>% glimpse()
```

```
## Rows: 4,269
## Columns: 12
## $ no_of_dependents      <dbl> 2, 0, 3, 3, 5, 0, 5, 2, 0, 5, 4, 2, 3, 2, 1, ~
## $ education            <chr> "Graduate", "Not Graduate", "Graduate", "Grad~
## $ self_employed        <chr> "No", "Yes", "No", "No", "Yes", "Yes", "No", ~
## $ income_annum         <dbl> 9600000, 4100000, 9100000, 8200000, 9800000, ~
## $ loan_amount          <dbl> 29900000, 12200000, 29700000, 30700000, 24200~
## $ loan_term            <dbl> 12, 8, 20, 8, 20, 10, 4, 20, 20, 10, 2, 18, 1~
## $ cibil_score          <dbl> 778, 417, 506, 467, 382, 319, 678, 382, 782, ~
## $ residential_assets_value <dbl> 2400000, 2700000, 7100000, 18200000, 12400000~
## $ commercial_assets_value <dbl> 17600000, 2200000, 4500000, 3300000, 8200000,~
## $ luxury_assets_value   <dbl> 22700000, 8800000, 33300000, 23300000, 294000~
## $ bank_asset_value      <dbl> 8000000, 3300000, 12800000, 7900000, 5000000,~
## $ loan_status          <chr> "Approved", "Rejected", "Rejected", "Rejected~
```

Encabezado de la base de datos.

```
## # A tibble: 6 x 12
##   no_of_dependents education    self_employed income_annum loan_amount loan_term
```

```
##           <dbl> <chr>           <chr>           <dbl>           <dbl>           <dbl>
## 1           2 Graduate         No           9600000         29900000         12
## 2           0 Not Graduate     Yes          4100000         12200000         8
## 3           3 Graduate         No           9100000         29700000         20
## 4           3 Graduate         No           8200000         30700000         8
## 5           5 Not Graduate     Yes          9800000         24200000         20
## 6           0 Graduate         Yes          4800000         13500000         10
## # i 6 more variables: cibil_score <dbl>, residential_assets_value <dbl>,
## #   commercial_assets_value <dbl>, luxury_assets_value <dbl>,
## #   bank_asset_value <dbl>, loan_status <chr>
```

Se recodificaron las variables education y self_employed, para convertirlas en binarias.

```
df$education <- recode(df$education, "Graduate"=1, "Not Graduate"=0)
df$self_employed <- recode(df$self_employed, "No"=0, "Yes"=1)
```

Me aseguro de que la variable de salida loan_status sea tratada como factor.

```
df <- df %>%
  mutate(loan_status = relevel(as.factor(loan_status), "Approved", "Rejected"))
```

Me aseguro que la base quedó como la necesito.

```
head(df)
```

```
## # A tibble: 6 x 12
##   no_of_dependents education self_employed income_annum loan_amount loan_term
##           <dbl>     <dbl>         <dbl>         <dbl>         <dbl>         <dbl>
## 1           2         1           0           9600000         29900000         12
## 2           0         0           1           4100000         12200000         8
## 3           3         1           0           9100000         29700000         20
## 4           3         1           0           8200000         30700000         8
## 5           5         0           1           9800000         24200000         20
## 6           0         1           1           4800000         13500000         10
## # i 6 more variables: cibil_score <dbl>, residential_assets_value <dbl>,
## #   commercial_assets_value <dbl>, luxury_assets_value <dbl>,
## #   bank_asset_value <dbl>, loan_status <fct>
```

Una vista a la distribución de la variable de salida.

```
## # A tibble: 2 x 2
##   loan_status      n
##   <fct>         <int>
```

```
## 1 Approved      2656
## 2 Rejected      1613
```

Son 2656 casoa aprovados y 1613 rechazados.

Eliminamos `loan_status`, `cibil_score`, `loan_term` y `self_employed` porque no contribuyen a la clasificación.

```
df1<- df %>% dplyr::select(-loan_status,-cibil_score,-loan_term,-self_employed)
```

Así queda la base con la que vamos a hacer los dendrogramas.

```
head(df1)
```

```
## # A tibble: 6 x 8
##   no_of_dependents education income_annum loan_amount residential_assets_value
##         <dbl>      <dbl>      <dbl>      <dbl>          <dbl>
## 1             2          1    9600000    29900000         2400000
## 2             0          0    4100000    12200000         2700000
## 3             3          1    9100000    29700000         7100000
## 4             3          1    8200000    30700000        18200000
## 5             5          0    9800000    24200000        12400000
## 6             0          1    4800000    13500000         6800000
## # i 3 more variables: commercial_assets_value <dbl>, luxury_assets_value <dbl>,
## #   bank_asset_value <dbl>
```

Calculamos las distancia euclideanas con 5 ligas: promedio, única, completa, Ward y Ward2..

```
dist_df <- dist(scale(df1), method = "euclidean") #dist_df es un vector, con las distancias entre cada
```

```
hc_ave <- hclust(dist_df, method = "average")
hc_single<-hclust(dist_df, method = "single")
hc_comp<-hclust(dist_df, method = "complete")
hc_ward<-hclust(dist_df, method = "ward.D")
hc_ward2<-hclust(dist_df, method = "ward.D2")
```

Se calcula la distancia cophenetic de cada liga.

```
d_ave <- cophenetic(hc_ave)
d_single <- cophenetic(hc_single)
d_comp<- cophenetic(hc_comp)
d_ward<- cophenetic(hc_ward)
d_ward2<-cophenetic(hc_ward2)
```

Comparamos la correlación de cada una de las ligas.

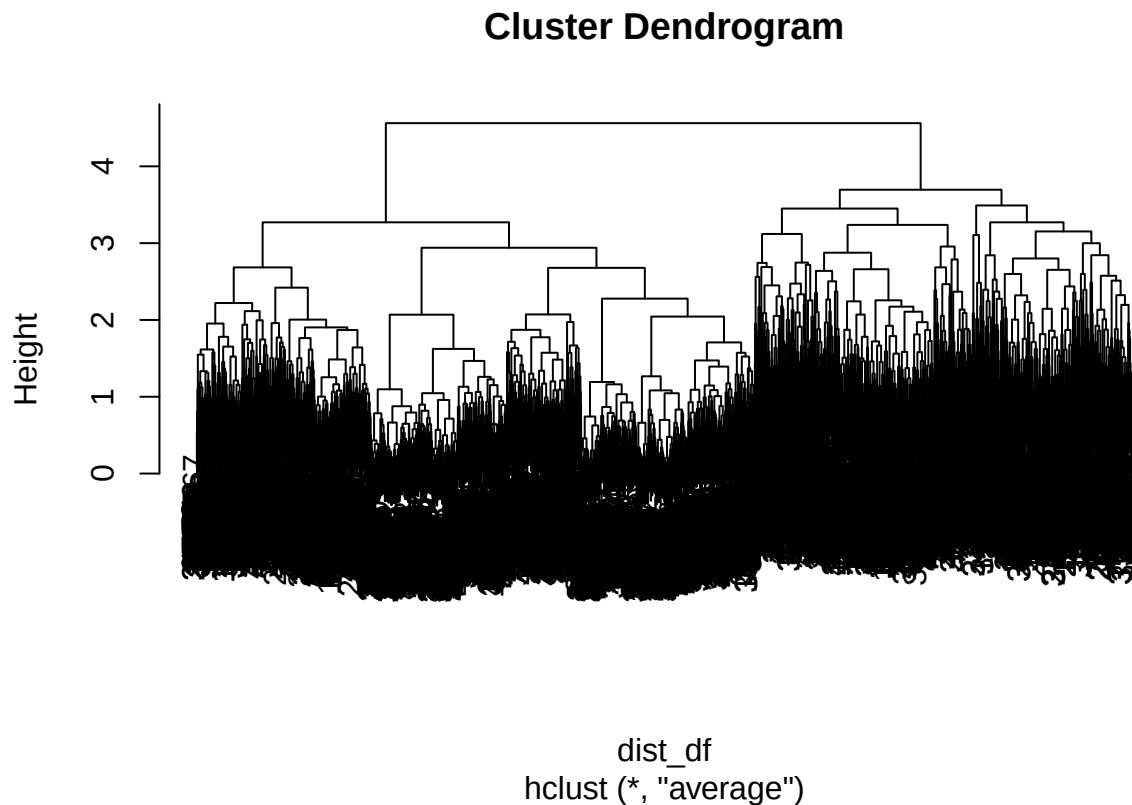
```
cbind(cor(dist_df,d_ave),cor(dist_df,d_single),cor(dist_df,d_comp),cor(dist_df,d_ward),cor(dist_df,d_wa
```

```
##           [,1]      [,2]      [,3]      [,4]      [,5]  
## [1,] 0.6412293 0.3168403 0.6483186 0.5751081 0.6114007
```

La liga que más correlación muestra es la liga completa, muy cerca de la liga promedio.

Graficamos los dendrogramas.

```
plot(hc_ave)
```



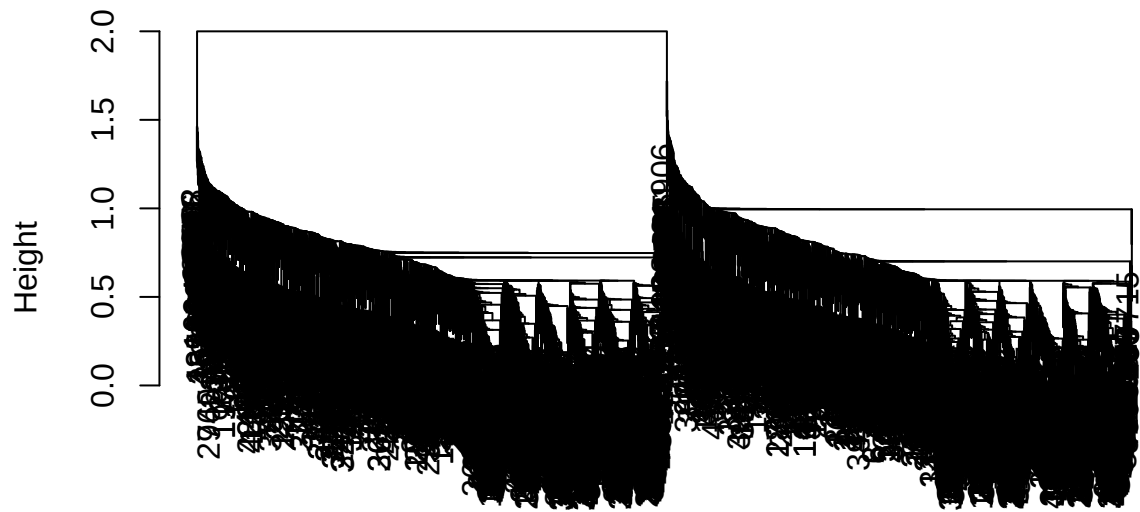
La liga promedio se ve muy poco claro.

Probamos la liga simple.

Se alcanza a ver dos grupos grandes. La muestra tiene muchas observaciones, por eso en la base se ven solo manchas negras, sin embargo, se pueden distinguir dos grupos.

```
plot(hc_single)
```

Cluster Dendrogram

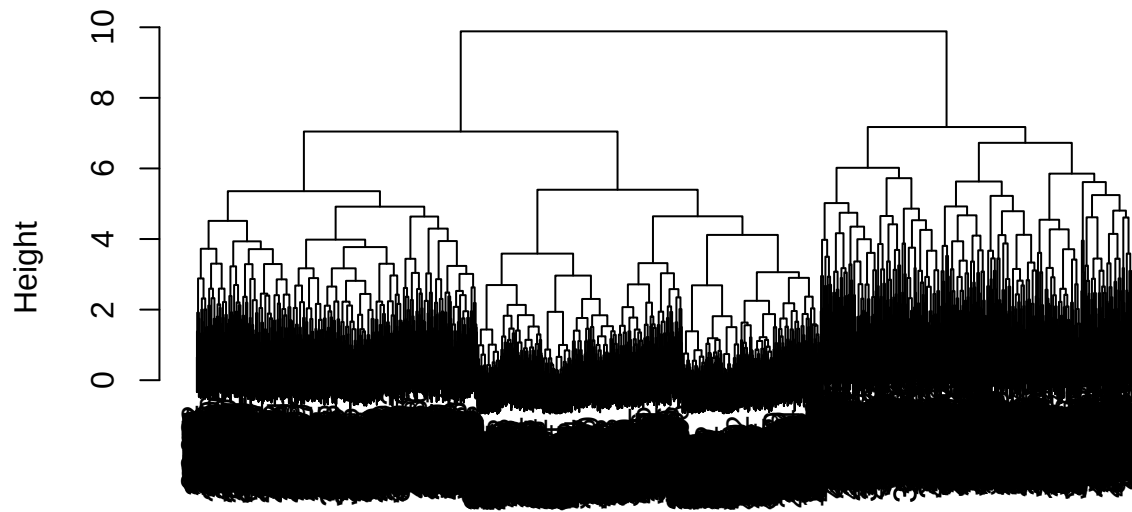


```
dist_df
hclust (*, "single")
```

Liga completa

```
plot(hc_comp)
```

Cluster Dendrogram

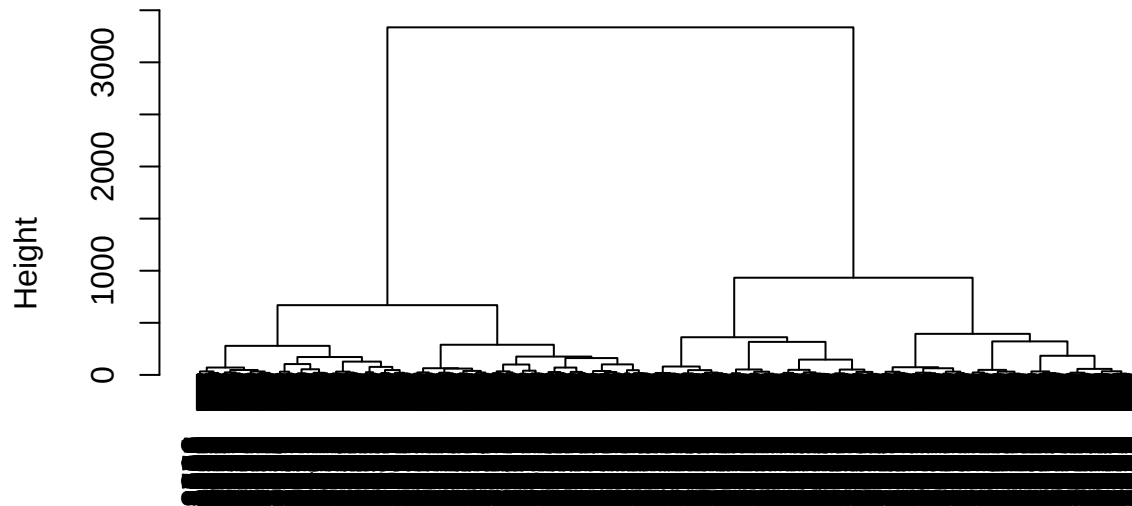


```
dist_df
hclust (*, "complete")
```

Con Ward parece que es clara la separación en dos grupos a la altura de 1000.

```
plot(hc_ward)
```

Cluster Dendrogram

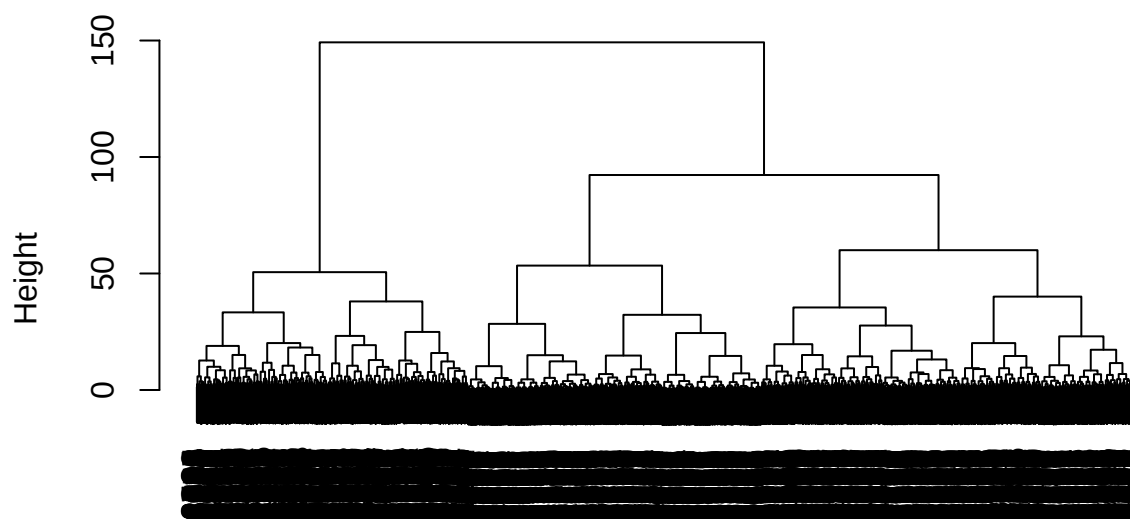


```
dist_df  
hclust (*, "ward.D")
```

Con Ward2 se identifican dos grupos a una altura de 100 y tres poco arriba de 50.

```
plot(hc_ward2)
```

Cluster Dendrogram



```
dist_df
hclust (*, "ward.D2")
```

Para comparar usamos el método agnes.

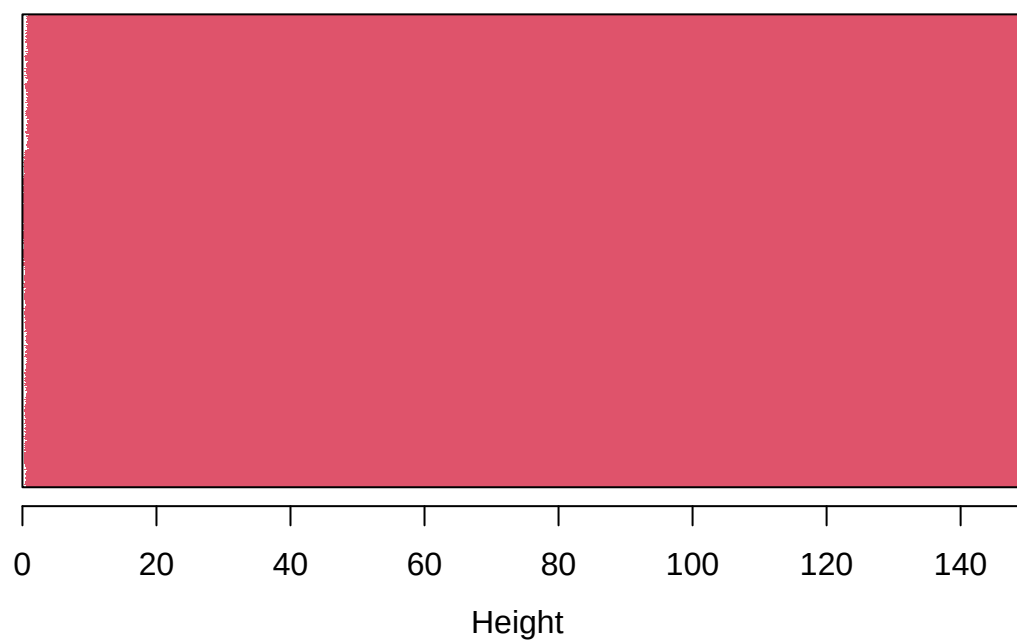
```
ac_metric <- list(
  complete_ac = agnes(df1, metric = "euclidean", method = "complete")$ac,
  average_ac = agnes(df1, metric = "euclidean", method = "average")$ac,
  single_ac = agnes(df1, metric = "euclidean", method = "single")$ac,
  ward_ac = agnes(df1, metric = "euclidean", method = "ward")$ac
)

ac_metric
```

```
## $complete_ac
## [1] 0.9399648
##
## $average_ac
## [1] 0.8717354
##
## $single_ac
## [1] 0.7275024
##
## $ward_ac
## [1] 0.9960268
```

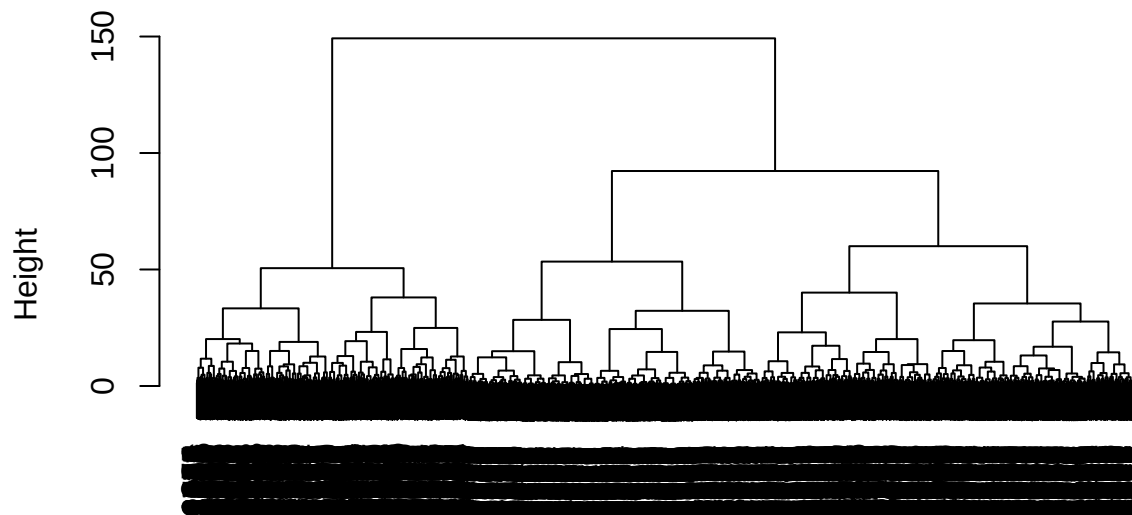
```
plot(agnes(df1, metric = "euclidean", method = "ward"))
```

Banner of `agnes(x = df1, metric = "euclidean", method = "ward.D2")`



Agglomerative Coefficient = 1

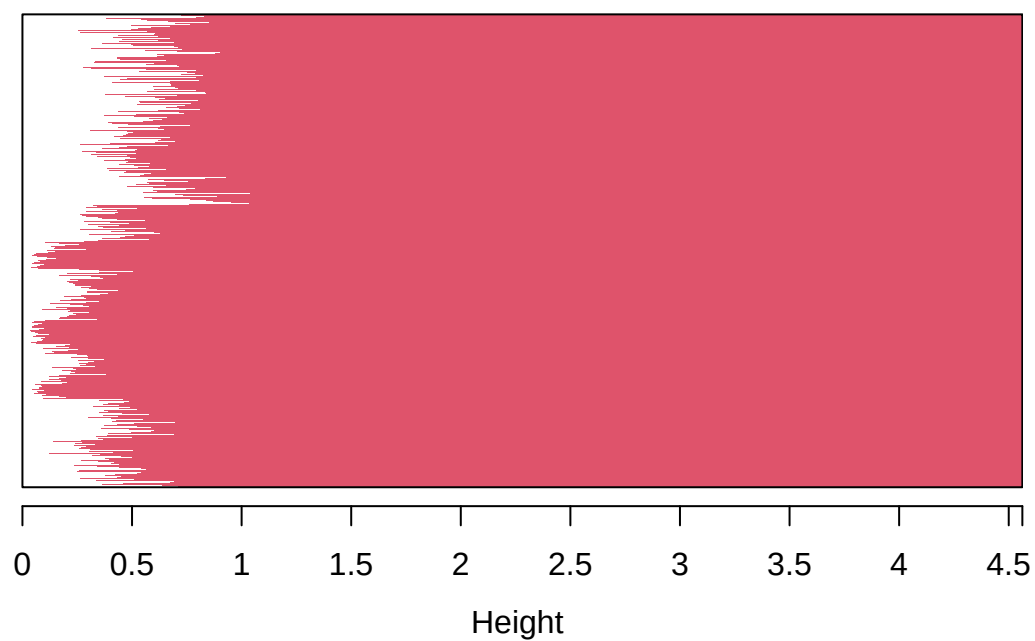
Dendrogram of `agnes(x = df1, metric = "euclidean", method = "ward`



df1
Agglomerative Coefficient = 1

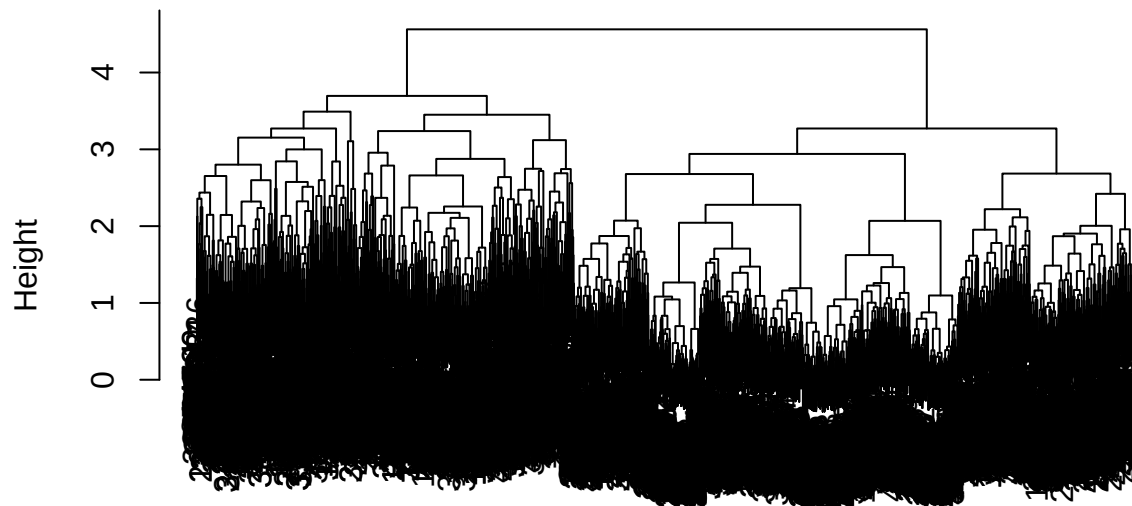
```
plot(agnes(df1, metric = "euclidean", method = "average"))
```

Banner of `agnes(x = df1, metric = "euclidean", method = "ave`



Agglomerative Coefficient = 0.87

Dendrogram of `agnes(x = df1, metric = "euclidean", method = "average")`

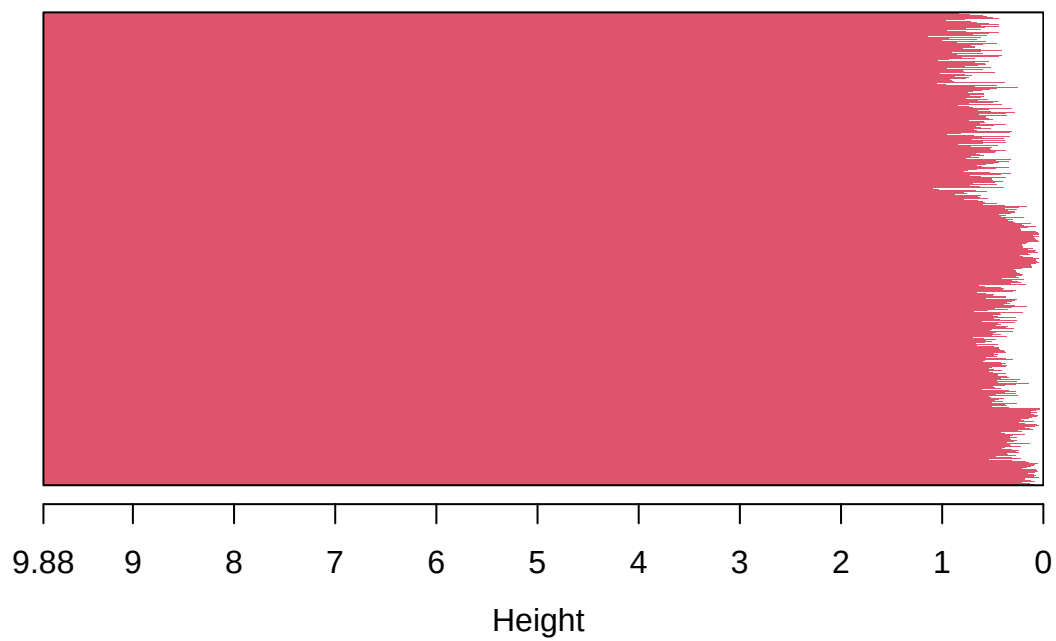


df1
Agglomerative Coefficient = 0.87

Los resultados se ven similares.

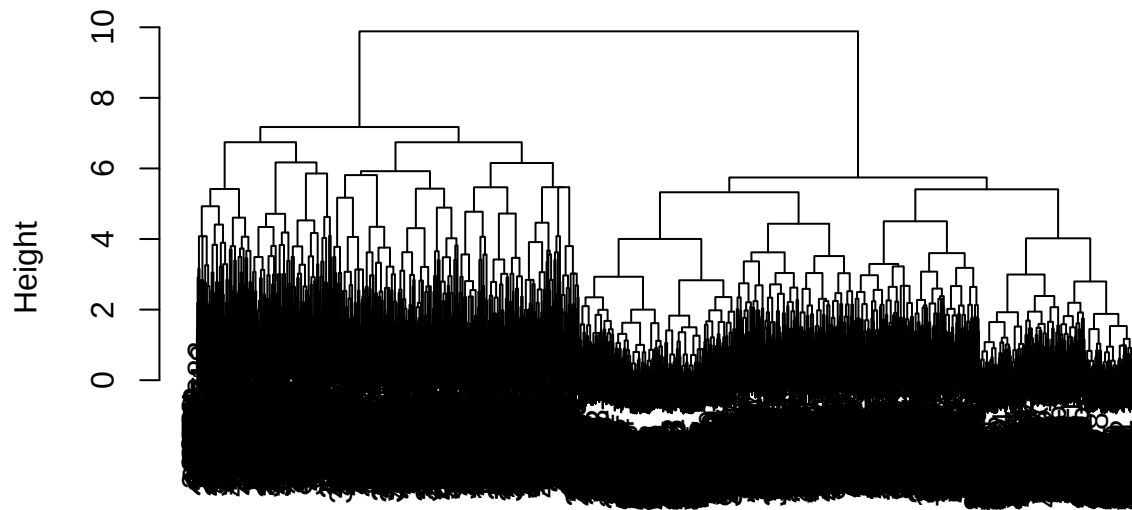
```
plot(diana(df1, metric = "euclidean"))
```

Banner of `diana(x = df1, metric = "euclidean")`



Divisive Coefficient = 0.93

Dendrogram of `diana(x = df1, metric = "euclidean")`



df1

Divisive Coefficient = 0.93

El Método de dendrogramas no es muy claro. Se pueden observar entre dos y tres grupos. La liga es la que mejor permite separar los grupos es Ward. En este caso, sabemos que hay dos grupos, por ello podemos seleccionar Ward.